

E.3 / PAPER 1 / QUESTION 19

29 April 2025 / TZ3 (file label) / HL Paper 1A / original Q35 / 2225-6719

- 19.** Thorium-232 ($^{232}_{90}\text{Th}$) is an unstable nuclide. It decays by emitting an alpha particle followed by a beta minus (β^-) particle.

What is the number of protons and neutrons in the final nuclide?

	Number of protons	Number of neutrons
A.	87	139
B.	87	141
C.	89	139
D.	89	141